2013-2014 – Master 2 – Macro I

Lecture 9 : Asset Values

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Disclaimer

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1. Model

- Consider an asset which pays a dividend d_t
- Value at t *before dividends* is V_t.
- Assume that I can get a rate of return elsewhere equal to r.
- If I sell my asset now and invest the proceeds at rate r, I will get $(1+r)V_t$ tomorrow.
- If I hold it, I will get $d_t + V_{t+1}$.
- Therefore the arbitrage condition is

$$V_t = d_t + \frac{V_{t+1}}{1+r}. \tag{1}$$

- Equivalently we can write :

$$rV_t = (1+r)d_t + (V_{t+1} - V_t). \tag{2}$$

1. Model

- That is :

  `Rate of Return x Value of the asset = Dividends + Capital Gains.`

- Note that for risk neutral agents, the formula extends to risky investments :

$$rV_t = (1 + r)d_t + E_t(V_{t+1} - V_t).$$

- That is :

  `Rate of Return x Value of the asset = Dividends + Expected Capital Gains`

1. Model

- An equation like (1) can be solved by *forward integration*

$$V_t = \sum_{i=t}^{T} \frac{1}{(1+r)^{i-t}} d_i + \frac{V_{T+1}}{(1+r)^{T+1-t}}. \tag{3}$$

- Assume that the following sequence exists :

$$F_t = \sum_{i=t}^{+\infty} \frac{1}{(1+r)^{i-t}} d_i \tag{4}$$

- Then if $r > 0$ F_t is the only non explosive solution to (3).
- If we eliminate explosive solutions, then we have a unique one which is called the *fundamental* value of the asset.
- Furthermore, let $\tilde{V}_t$ be any other solution.
- Let $B_t = \tilde{V}_t - F_t$.
- It must be that

$$B_{t+1} = (1+r)B_t.$$

- Therefore, any solution is the sum of the "fundamental" and a "bubble" which explodes at rate r.

1. Model

- Remark : In fact the bubble can be stochastic and only need to explode at rate r in expectations. That is, up to a multiplicative factor $(1+r)$ the bubble can be any martingale.
- Assume that the bubble can burst with probability p and that agents are risk neutral.
- Arbitrage requires $E_t B_{t+1} = (1+r)B_t$
- Now (with an abuse of notation as I denote the stochastic variable and the realization by the same symbol B_{t+1})

$$E_t B_{t+1} = (1-p)B_{t+1} + p \times 0$$

- Therefore $(1-p)B_{t+1} + p \times 0 = (1+r)B_t$, so that

$$\begin{aligned} B_{t+1} &= \frac{1+r}{1-p} B_t \\ &\approx (1+r+p)B_t \quad \text{when it does not burst} \end{aligned}$$

1. Model

- Remark : Any present discounted value of an income (dividend) flow can be rewritten in a recursive fashion. For example if

$$V_t = \sum_{i=t}^{+\infty} \frac{1}{(1+r)^{i-t}} d_i, \tag{5}$$

- This can be written

$$V_t = d_t + \frac{1}{1+r} V_{t+1}. \tag{6}$$

Then this can be again interpreted in a dividend plus capital gains fashion :

$$rV_t = (1+r)d_t + V_{t+1} - V_t. \tag{7}$$

2. Continuous Time

Pricing the asset

- Suppose an asset yields a flow of dividends per unit of time equal to $x(t)$.
- Suppose I can get a return r per unit of time elsewhere.
- Let $V(t)$ be the value of the asset at t.
- Suppose I sell it and invest in the market over a small time interval dt.
- Then at $t + dt$ I will have $V(t)(1 + r.dt)$.
- If instead I hold the asset, then at $t + dt$ I will have $x(t).dt + V(t + dt)$.
- Hence
$$x(t).dt + V(t + dt) = V(t)(1 + r.dt), \tag{8}$$
- Neglecting second order terms, $V(t + dt) \approx V(t) + \dot{V}(t)dt$
- so that (8) writes
$$rV(t) = x(t) + \dot{V}(t)$$

2. Continuous Time

Pricing the asset

$$rV(t) = x(t) + \dot{V}(t)$$

- The LHS is the rate of return (per unit of time) times the value of the asset.
- The RHS is the dividend per unit of time, plus the expected capital gain per unit of time.
- The equation is the same as (2), except that the arbitrage takes place over a small time interval and that all is expressed per unit of time.
- In continuous time the fundamental is
$$F_t = \int_t^{+\infty} e^{-r(v-t)} x(v) dv$$
- And a bubble is such that
$$\dot{B} = rB \quad \text{that is} \quad B_t = B_0 e^{rt}.$$

2. Continuous time
Exogenous shocks

- Assume risk neutrality
- Assume that we have an asset whose value at date t is $V(t)$, pays dividends $x(t)$ and that this asset can exogenously be transformed into another asset.
- This other asset date t value is denoted $\widetilde{V}(t)$.
- Example : Consider that the asset is a car, whose value is V. With some probability, the owner has an accident and the car turns into a broken car, whose value is denoted $\widetilde{V}$
- The change of nature (the car accident) is Poisson distributed, with intensity $\lambda > 0$

2. Continuous time

Exogenous shocks

- Note : for a Poisson process with intensity $\lambda > 0$, the probability of having k events in time interval h is $\frac{e^{-\lambda h}(\lambda h)^k}{k!}$
- Note : when h goes to 0,
  - Probability of observing 0 events in a time interval h $= e^{-\lambda h} \approx 1 - \lambda h$
  - Probability of observing 1 events in a time interval h $= e^{-\lambda h}\lambda h \approx \lambda h$
  - Probability of observing more than events in a time interval h ≈ 0

2. Continuous time

Exogenous shocks

- How to write the asset valuation equation ?
- Let $V(t)$ be the value of the asset at t.
- Suppose I sell it and invest in the market over a small time interval dt.
- Then at $t + dt$ I will have $V(t)(1 + r.dt)$.
- If instead I hold the asset, then at $t + dt$ I will have

$$x(t).dt + (1 - \lambda dt)V(t + dt) + \lambda dt \widetilde{V}(t + dt)$$

- Hence, by arbitrage

$$V(t)(1 + r.dt) = x(t).dt + (1 - \lambda dt)V(t + dt) + \lambda dt \widetilde{V}(t + dt)$$

2. Continuous time

Exogenous shocks

$$V(t)(1+r.dt) = x(t).dt + (1-\lambda dt)V(t+dt) + \lambda dt\widetilde{V}(t+dt) \ \ (\star)$$

- First order Taylor expansion :

$$\begin{cases} V(t+dt) & = & V(t) + \dot{V}(t)dt \\ \widetilde{V}(t+dt) & = & \widetilde{V}(t) + \dot{\widetilde{V}}(t)dt \end{cases}$$

- $(\star)$ becomes

$$\begin{aligned} V(t) + rV(t)dt \ &= \ x(t)dt + V(t) + \dot{V}(t)dt - \lambda V(t)dt - \dot{V}(t)(dt)^2 \\ &+ \ \lambda\widetilde{V}(t)dt - \dot{\widetilde{V}}(t)(dt)^2 \end{aligned}$$

- Simplify the blue terms, neglect the red ones gives

$$rV(t) = x(t) + \lambda\left(\widetilde{V}(t) - V(t)\right) + \dot{V}(t)$$